



HÖHERE TECHNISCHE BUNDESLEHRANSTALT HOLLABRUNN  
COLLEGE of ENGINEERING

Information Technology

# DIPLOMA THESIS

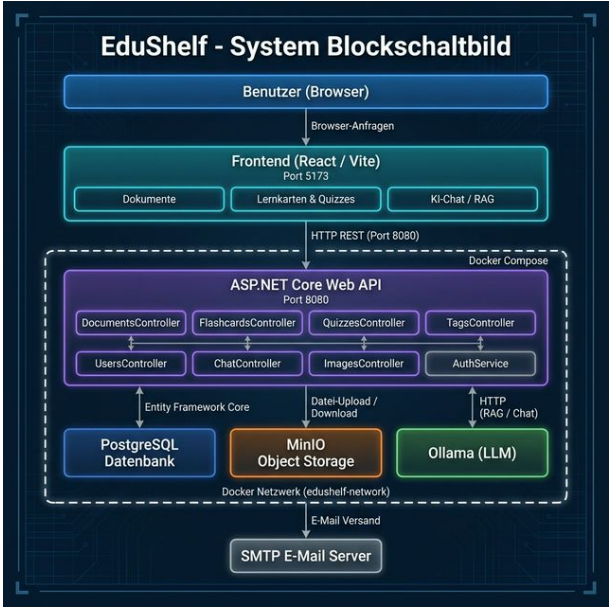
## Documentation

|                       |                                 |
|-----------------------|---------------------------------|
| Author(s)             | Elias Welt, Phillip Rubak,<br>, |
| From<br>Academic year | 2025/26                         |
| Topic                 | EduShelf                        |
| Co-operation partners | X-Works                         |

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| Task Description | This diploma project develops 'EduShelf', a web-based platform for managing educational content. It provides students and teachers with a centralized solution for organizing documents and leveraging AI-powered information retrieval (RAG – Retrieval Augmented Generation). The system integrates a modern web frontend, a backend for data processing and storage, and a dedicated AI server for intelligent chat functionality. |
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| Implementation | The frontend is built with React for a responsive, user-friendly interface. The .NET backend manages data storage and processing, while a Python/Ollama-based AI server handles LLM operations for the chat. Docker containerization ensures easy deployment and scalability. |
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| Results | The result is a fully functional 'EduShelf' platform featuring user authentication, document upload and indexing, and an interactive AI chat. Users can manage their document libraries and obtain instant answers from their study materials, demonstrating the effective integration of modern web technologies and AI in education. |
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| <p>Typische Grafik, Foto etc.<br/>(mit Erläuterung)</p> |  <p>The diagram, titled 'EduShelf - System Blockschaubild', illustrates the system architecture. At the top, a 'Benutzer (Browser)' box connects to a 'Frontend (React / Vite)' box (Port 5173) via 'Browser-Anfragen'. The Frontend contains sub-components for 'Dokumente', 'Lernkarten &amp; Quizzes', and 'KI-Chat / RAG'. The Frontend connects to an 'ASP.NET Core Web API' box (Port 8080) via 'HTTP REST (Port 8080)'. The Web API is part of a 'Docker Compose' environment and contains several controllers: 'DocumentsController', 'FlashcardsController', 'QuizzesController', 'TagsController', 'UsersController', 'ChatController', 'ImagesController', and 'AuthService'. Below the Web API, there are three main components: 'PostgreSQL Datenbank' (connected via 'Entity Framework Core'), 'MinIO Object Storage' (connected via 'Datei-Upload / Download'), and 'Ollama (LLM)' (connected via 'HTTP (RAG / Chat)'). These three components are part of a 'Docker Netzwerk (edushelf-network)'. Finally, the 'SMTP E-Mail Server' is connected to the network via 'E-Mail Versand'.</p> |
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| Participation in competitions<br>Awards |  |
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| <p>Accessibility of<br/>final project thesis</p> | <p>HTL Hollabrunn<br/>Anton Ehrenfriedstraße 10<br/>2020 Hollabrunn</p> |
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| Approval (Date / Signature) | Examiner/s | Head of Department / College |
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